FIITJEE

JEE(Advanced)-2019

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – I

PAPER-2

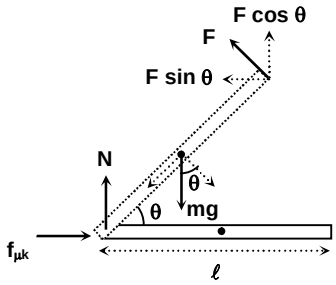
Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C, D**	19.	**A, B, C**	37.	**C, D**
2.	**A, B**	20.	**A, C**	38.	**B, D**
3.	**C, D**	21.	**C, D**	39.	**A, B, C**
4.	**A, B, C**	22.	**A, B, C**	40.	**A, B, C, D**
5.	**A, B, C, D**	23.	**A, C**	41.	**A, B, D**
6.	**A, C**	24.	**C, D**	42.	**A, B, C, D**
7.	**C**	25.	**B**	43.	**D**
8.	**A**	26.	**D**	44.	**C**
9.	**A**	27.	**B**	45.	**B**
10.	**B**	28.	**A**	46.	**D**
11.	**00002.00**	29.	**00046.06**	47.	**00000.25**
12.	**00005.00**	30.	**00015.50**	48.	**00000.50**
13.	**00001.00**	31.	**00004.74**	49.	**00000.50**
14.	**00002.00**	32.	**00005.27**	50.	**00001.50**
15.	**00003.00**	33.	**00028.04**	51.	**00000.40**
16.	**00002.00**	34.	**01215.60**	52.	**00000.50**
17.	**00006.00**	35.	**00115.15**	53.	**00000.25**
18.	**00002.00**	36.	**02763.60**	54.	**00000.50**

Physics

PART – I

SECTION – A

1. Since the rod is raised slowly (quasi-statically), the entire system remains in equilibrium at any moment in time. Thus, the torque about the point of rotation of the rod is zero about any axis, and the net external forces are 0. $F_{fric} = F \sin\theta$



$$\tau = -Fl + \frac{mgl\cos\theta}{2} = 0 \Rightarrow F = \frac{mg\cos\theta}{2}$$

$N = mg - F\cos\theta$

The F sin θ force causes the rod's bottom end to slip in one direction. The friction force counters this force to stop the slipping

when $F_{fric} = \mu_s N \geq F\sin\theta$. Hence,

$$\mu_s \geq \frac{F\sin\theta}{mg - F\cos\theta} = \frac{\sin\theta\cos\theta}{2 - \cos^2\theta}$$

Maximum of μs can be found by setting its derivative with respect to theta equal to zero:

$$\frac{d\mu}{d\theta} = \frac{\left(2-\cos^2\theta\right)\left(2\cos^2\theta - 1\right) - 2\cos^2\theta\left(1-\cos^2\theta\right)}{\left(2-\cos^2\theta\right)^2} = 0$$ This gives $\sin\theta = \frac{1}{\sqrt{3}}$ and $\mu_{min} = \frac{\sqrt{2}}{4}$

2. First consider the situation immediately after the perfectly inelastic collision between B and C but before A has begun to slip onto C; label this as the initial configuration "i." Clearly $v_{Ai} = v$ and $v_{Bi} = v_{Ci} = v/2$ by momentum conservation. On the other hand, in the end (labeled "f") all three masses have the same final speed, so that $v_{Af} = v_{Bf} = v_{Cf} = 2v/3$ by again applying momentum conservation. The sum of the kinetic energies of the boards thus changes by

$\Delta K = (K_{Af} + K_{Bf} + K_{Cf}) - (K_{Ai} + K_{Bi} + K_{Ci}) = -(mv^2/12)$ …(1)

where m is the mass of each board. This net loss in mechanical energy is the result of the non conservative "internal work" done by friction, $W_{NC} = -\int f dx_A + \int f dx_C = -\int_0^L f d\left(x_A - x_C\right)$ …(2)

where the equal and opposite internal force of friction between blocks A and C is given by

$f = \mu_k N = \mu_k\, mg(x_A - x_C)/L$ …(3)

with L the length of each board. Here, x_A and x_C are the rightward displacements of blocks A and C from their initial positions; the ratio $(x_A - x_C)/L$ thus gives the fraction of block A, which is on top of block C, and hence the ratio of the normal force N of A on C to the weight mg of A. Substitute Eq. (3) into Eq. (2) to find

$$W_{NC} = \frac{\mu_K mg}{L}\frac{L^2}{2}$$ …(4)

It is worth pausing to note that it would have been extremely difficult to attempt to separately calculate the two frictional work terms in the middle expression in Eq. (2)! Finally equate the right-hand sides of Eqs. (1) and (4) to obtain the answer,

$$L = \frac{v^2}{6\mu_k g}$$

3. $e = 1 = \dfrac{v_1 + v_2}{2\sqrt{2g\ell}}$ …(i)

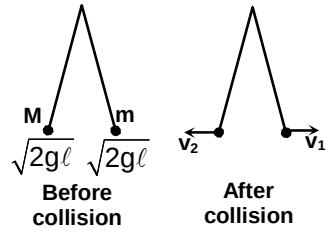
Momentum conservative

$M\sqrt{2g\ell} - m\sqrt{2g\ell} = mv_1 - Mv_2$ …(ii)

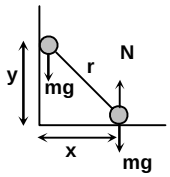
Solving (i) and (ii)

$v_1 = \left(\dfrac{3M-m}{M+m}\right)\sqrt{2gL} \geq \sqrt{5gl}$

$\Rightarrow \dfrac{M}{m} \geq \dfrac{\sqrt{5}+\sqrt{2}}{3\sqrt{2}-\sqrt{5}} = 1.82 = 1.82$



4. Since the normal force on the top mass is zero and the horizontal acceleration of the top mass is zero at the instant it loses con-tact with the wall, the tension in the rod at that moment must be zero. Thus, free-body diagrams for the two masses are as sketched below Therefore, the top mass has downward velocity v = – dy/dt and acceleration $g = -d^2y/dt^2$,



While the bottom mass has rightward velocity u = dx/dt and zero acceleration. But $y = (r^2-x^2)^{1/2}$.

$$v = -\frac{dy}{dt} = \frac{x}{\sqrt{r^2-x^2}}\frac{dx}{dt} = \frac{xu}{y}$$

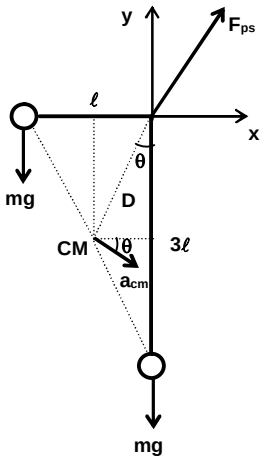
$$g = -\frac{d^2y}{dt^2} = \frac{u}{y}\frac{dx}{dt} - \frac{xu}{y^2}\frac{dy}{dt} = \frac{u^2}{y} + \frac{xuv}{y^2}$$

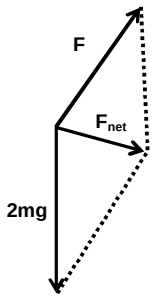
$$mg(r-y) = \frac{1}{2}m\left(u^2+v^2\right)$$

$$\Rightarrow 2g(r-y) = u^2\left(1+\frac{x^2}{y^2}\right) = \frac{u^2r^2}{y^2} = gy$$

$$y = \frac{2}{3}r \Rightarrow u = \sqrt{\frac{8gr}{27}}$$

5. According to Newton's third law, the force F_{sp} that the system (s) exerts on the pivot (p) is opposite to the force F_{ps} that the pivot exerts on the system, $F_{sp} = -F_{ps}$. Then, we can use Newton's second law to first deduce Fps and so to obtain the required force F_{sp}. The physical situation is represented in the figure below.





Applying Newton's second law to the system:

$F_{net} = M\, a_{cm}$, (1)

where F_{net} indicates the sum of all external forces on the system, a_{cm} indicates the acceleration of the center of mass of the system, and M is the total mass of the system that is M = 2m because the mass of the "light rod" is neglected. In the present

case Fnet is the sum of the gravitational force on the two spheres and the force Fps that exerts the pivot:

$F_{net} = M\, g + F_{ps}$. (2)

The force Fps can be deduced from Eqs. (1) and (2):

$F_{ps} = M\, a_{cm} - M\, g$. (3)

Then, to obtain F_{ps} it is necessary to have a_{cm}. We note that just after the system is released, the acceleration of the center of mass is tangent to the circular trajectory that the center of mass will follow during its oscillation around the pivot (the pivot is the suspension point and the system will behave as a physical pendulum). It allows us to relate the magnitude of this tangential acceleration with the magnitude of the angular acceleration a of the system:

$a_{cm} = \alpha D$, (4)

where D is the distance between the pivot and the center of mass. From the figure,

$$D = \frac{\sqrt{10}}{2} l \quad (5)$$

Equation (4) is only valid just after the system is released at t = 0, because for t > 0 the system will also have a centripetal acceleration pointing to the pivot. The angular acceleration a is obtained by using Newton's second law in "its rotational version." We have:

$$\tau_{net} = I\alpha \quad (6)$$

where τ_{net} has to be measured with respect to an axis that passes through the pivot and is perpendicular to the vertical plane when the system is contained. Since the weight of the upper mass is the only force that produces a torque, the magnitude of this torque can be found as:

$\tau_{net} = mgl$ (7)

Here I is the moment of inertia of the system with respect to the same axis:

$I = 10\, m^2$ (8)

where we have neglected the moment of inertia of the two arms that the light rod forms, due to the assumption $m_{rod} << m$. Combining Eqs. (6), (7), and (8) we obtain

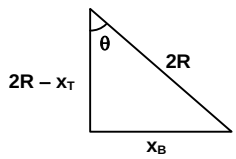
$\alpha = g/10l$ (9)

$$a_{cm} = \frac{g}{10\sqrt{2}} \quad (10)$$

$$\overrightarrow{a_{cm}} = \left(\frac{3}{10}i - \frac{1}{20}j\right)\vec{g}$$

$$\overrightarrow{F_{ps}} = 2mg\left(\frac{3}{20}i - \frac{1}{20}j\right)g - 2mg(-j) = mg\left(\frac{3}{10}i + \frac{19}{20}j\right)$$

6. Consider a moment when the center of top the cylinder has moved down a distance x_T and the center of the bottom cylinder has moved to the right a distance x_B. At this moment, the cylinders are moving with speeds v_T and v_B, respectively. From Fig. , v_T and v_B, respectively. From Fig.,



$x_T = 2R(1-\cos\theta)$

Also, energy conservation gives:

$$m\left(\frac{v_B^{\,2} + v_T^{\,2}}{2}\right) = mgx_T = 2mgR(1-\cos\theta)$$

$$x_B^{\,2} + (2R - x_T)^2 = (2R)^2$$

Differentiating w.r.t. time

$v_T = v_B x_B/(2R - x_T) = v_B \tan\theta$.

Eliminating v_T from the energy conservation equation gives $v_B^2 = 4gR(1 - \cos\theta)/(1 + \tan^2\theta)$. Setting the derivative with respect to θ equal to zero gives

$\cos\theta_m = 2/3$

for the angle θ_m at which v_B is maximum. At this point, the speed of the bottom cylinder is $v_B = \sqrt{\frac{16gR}{27}}$.

Mathematically, it appears that the speed decreases after θ_m. However, for this to happen, the acceleration and, therefore, the horizontal force must be directed to the left (negative) for angles greater than θ_m. But the contact force on the bottom cylinder by the top cylinder cannot be directed to the left. Therefore, the bottom cylinder loses contact with the top one and moves off at the maximum speed of $v_B = \sqrt{\frac{16gR}{27}}$

7. Basic concept of collision

8. Taking torque about point A

$$N\sqrt{\ell^2-(\ell-h)^2} = f_s(\ell-h)$$

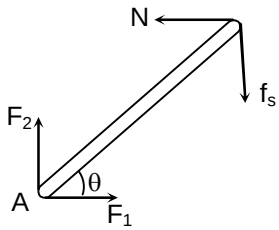
$$\Rightarrow f_s = \frac{N\sqrt{\ell^2-(\ell-h)^2}}{\ell-h} \le \mu N$$

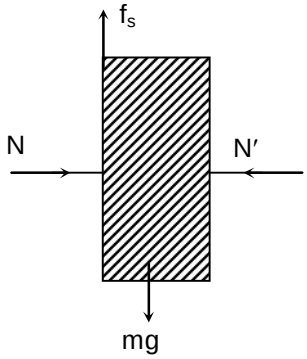
$$\Rightarrow \ell^2-(\ell-h)^2 \le \mu^2(\ell-h)^2$$

$$\Rightarrow \ell^2 \le (\mu^2+1)(\ell-h)^2$$

$$\Rightarrow \frac{\ell}{\sqrt{\mu^2+1}} \le (\ell-h)$$

$$\Rightarrow h \le \ell\left[1-\frac{1}{\sqrt{1+\mu^2}}\right] \Rightarrow h_{max} = 2m$$





$f_s = 3$ N

$N = 4N$

$F_H = 5$ N.

9. (P) $a < \frac{g}{3\sqrt{3}} \Rightarrow$ 3, 4

(Q) $\frac{g}{\sqrt{3}} < a < \frac{g}{3\sqrt{3}} \Rightarrow$ 2

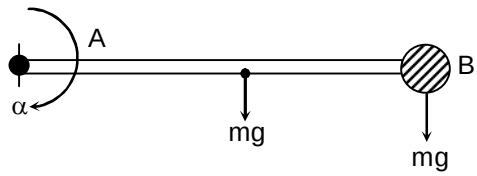
(R) 5

(S) $a > \frac{g}{\sqrt{3}} \Rightarrow 1$

10. $$Mg\frac{\ell}{2} + mg\ell = \left(\frac{M\ell^2}{3} + m\ell^2\right)\alpha$$

$$\alpha = \left(\frac{\frac{M}{2}+m}{\frac{M}{3}+m}\right)\frac{g}{\ell} = \frac{40}{3}\ \text{rad/s}^2$$

$$mg + F_1 = m\left(\frac{40}{3}\times 1\right)$$



i.e. $F_1 = \frac{40}{3} - 10 = \frac{10}{3}$

$Mg + mg - F_H = M\left(\frac{20}{3}\right) + m\left(\frac{40}{3}\right)$

$= 40 + \frac{40}{3}$

$70 - F_H = \frac{160}{3}$

$F_H = 70 - \frac{160}{3} = \frac{50}{3}$

SECTION – D

11. $a_1 \cos[90° - (\phi - \alpha)] = a_2 \sin\alpha$

$a_1 \sin(\phi - \alpha) = a_2 \sin\alpha$ …(i)

Conservation has no acceleration x direction

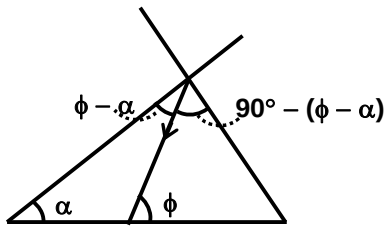
$\Rightarrow Ma_2 = ma_1 \sin\phi$ …(ii)

From (i) and (ii)

$\Rightarrow \frac{m}{M}\sin\phi = \frac{\sin(\phi - \alpha)}{\sin\alpha}$

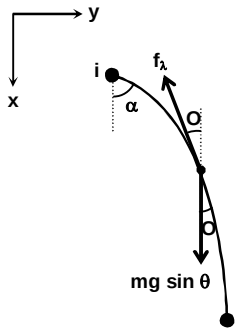
$\Rightarrow \frac{m}{M} = \frac{\tan\phi - \tan\alpha}{\tan\alpha}$

$\frac{m}{M} = \frac{\tan 60 - \tan 30}{\tan 30} = 2$



12. $\frac{\sqrt{5}v_o}{2}$ (Apply conservation of energy)

13. Because the block does not accelerate off the plane surface, the normal force N that the surface exerts on the block (of mass m) must balance the component of the block's weight perpendicular to the plane, so that N =mg cos θ, and thus the kinetic frictional force is $f_k = \mu_k$ mg cos θ. On the other hand, the following sketch shows the path of the block (between its initial position "i" and final position "f ") and the components of the forces acting on it in the plane of the incline. I have chosen coordinates with x down the incline and y perpendicular to it. A tangent to the block's path makes angle f with respect to the x-direction, with $\varphi_i = \alpha$. (It is interesting to note that φ must be zero because the block's final speed and hence its final centripetal acceleration must approach zero, which can only happen if the frictional force points up the plane, opposite the weight component down the plane.)



Newton's second law in the x-direction, after dividing both sides by the block's mass, becomes

$\frac{dv_x}{dt} = g\sin\theta - \mu_k g\cos\theta\cos\phi$ (1)

where u is the block's speed, with initial value u_0. Writing down the y-component of Newton's second law introduces sin ϕ, which leads to a mess. A better choice is to instead write down the tangential component of Newton's second law (noting that any direction other than x would give an independent equation),

$$\frac{dv}{dt} = g\sin\theta\sin\phi - \mu_k g\cos\theta \qquad (2)$$

Solve Eq. (1) for cos φ (which is the only time-dependent quantity on the right-hand side of these two equations) and substitute it into Eq. (2) to get

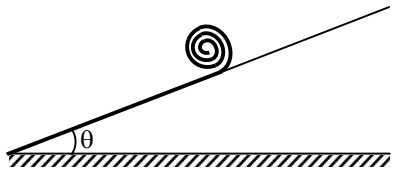
$$\mu_k \frac{dv}{dt} = g\cos\theta\left(\tan^2\theta - \mu_k{}^2\right) - \tan\theta\frac{dv_x}{dt}$$

after simplifying. Now integrate both sides over the time t from the initial to the final point to obtain

$$\mu_k\left(0 - v_o\right) = g\cos\theta\left(\tan^2\theta - \mu_k{}^2\right)t - \tan\theta\left(0 - v_o\cos\alpha\right)$$

$$t = \frac{v_o\left(\mu_k + \tan\theta\cos\alpha\right)}{g\cos\theta\left(\mu_k{}^2 - \tan^2\theta\right)}$$

14. The rolled part of the carpet is a cylinder, and its mass is proportional to x. Applying conservation of mechanical energy, $\left(1+\frac{1}{2}\right)\frac{1}{2}(\lambda x)\left(\frac{dx}{dt}\right)^2 = (\lambda x)g\frac{x\sin\theta}{2}$



$$\therefore \quad \left(\frac{dx}{dt}\right)^2 = v^2 = \frac{2gx\sin\theta}{3}$$

i.e. $a_{carpet} = \frac{g}{3}$.

For the cylinder,

$$\frac{3}{4}mv^2 = mgx\sin\theta$$

i.e. $a_{cyl} = \frac{2g}{3}$

$$\frac{t_1}{t_2} = \sqrt{\frac{a_{cyl}}{a_{carpet}}} = \sqrt{2}$$

15. Maximum trajectory area

Let θ be the angle at which the ball is thrown. Then coordinates are given by x = (v cos θ)t and y = $(v\sin\theta)t - \frac{gt^2}{2}$. The total time in air is 2(v sin θ)/g, so the area under the trajectory, A = ∫ydx, is

$$\int_0^{x_{max}} ydx = \int_0^{2v\sin\theta/g}\left((v\sin\theta)t - \frac{gt^2}{2}\right)(v\cos\theta dt) = \frac{2v^4}{3g^2}\sin^3\theta\cos\theta$$

Taking the derivative of this, we find that the maximum occurs when tan θ = √3 that is, when θ = 60°. The maximum area is then $A_{max} = \frac{\sqrt{3}V^4}{8g^2}$. Note that by dimensional analysis we know that the area, which has dimensions of distance squared, must be proportional to $\frac{v^4}{g^2}$.

16. Circling around a pole

Let F be the tension in the string. At the mass, the angle between the string and the radius of the dotted circle is θ = $\sin^{-1}$(r/R). In terms of θ, the radial and tangential F = ma equations are

$$F\cos\theta = \frac{mv^2}{R}, \text{ and } F\sin\theta = m\frac{dv}{dt} \qquad \ldots(i)$$

Dividing these two equations gives tan θ = (Rv)/v^2. Separating variables and integrating gives

$$\int_{v_0}^{v}\frac{dv}{v^2}=\frac{\tan\theta}{R}\int_0^t dt$$

$$\Rightarrow \frac{1}{v_0}-\frac{1}{v}=\frac{(\tan\theta)t}{R}\Rightarrow v(t)=\left(\frac{1}{v_0}-\frac{(\tan\theta)t}{R}\right)^{-1} \quad \ldots(ii)$$

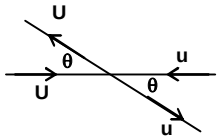
The speed v becomes infinite when

$$t=T=\frac{R}{v_0\tan\theta} \quad \ldots(iii)$$

This means that you can keep the mass moving the desired circle only up to time T. After that, it is impossible. (Of course, it will become impossible, for all practical purposes, long before v becomes infinite.) The total distance d = ∫vdt, is infinite, because this integral diverges (barely, like a log) as t approaches T.

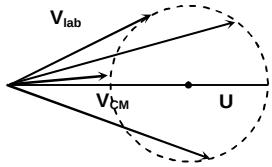
17. Let's figure out what the collision looks like in the CM frame. If M has initial speed V in the lab frame, then the CM moves with speed VCM = MV /(M +m). The speeds of M and m in the CM frame therefore equal, respectively,

$U = V - V_{CM} = \frac{mV}{M+m}$, and $u = \left|-V_{CM}\right| = \frac{MV}{M+m}$

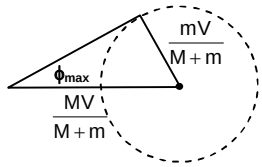


In the CM frame, the collision is simple. The particles keep the same speeds, but simply change their directions (while still moving in opposite directions), as shown in Figure.

The angle θ is free to have any value. This scenario clearly satisfies conservation of energy and momentum, so it must be what happens. The important point to note is that since θ can have any value, the tip of the U velocity vector can be located anywhere on a circle of radius U. If we then shift back to the lab frame, we see that the final velocity of M with respect to the lab frame, V_{lab}, is obtained by adding VCM to the vector U, which can point anywhere on the dotted circle in Figure.



A few possibilities for V_{lab} are shown. The largest angle of deflection is obtained when V_{lab} is tangent to the dotted circle, in which case we have the situation shown in Figure.

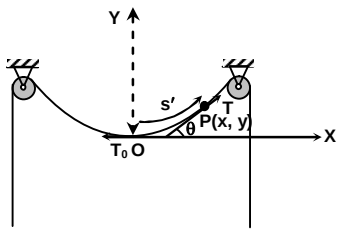


The maximum angle of deflection, φ_{max}, is therefore given by

$$\sin\phi_{max}=\frac{U}{V_{CM}}=\frac{mV/(M+m)}{MV/(M+m)}=\frac{m}{M}$$

If M < m, then the dotted circle passes to the left of the left vertex of the triangle. This means that φ can take on any value. In particular, it is possible for M to bounce directly backward.

18. s′ = the arc length OP.



$$T\cos\theta=T_0 \quad \ldots(1)$$

$$T\sin\theta=\lambda(s')g \quad \ldots(2)$$

$$\therefore \quad \tan\theta=\frac{dy}{dx}=\frac{dy}{dx}=\left(\frac{\lambda g}{T_0}\right)s' \quad \ldots(3)$$

Further: $\left.\begin{aligned} dy &= ds'\sin\theta \\ dx &= ds'\cos\theta \end{aligned}\right\}$ …(4)

Substituting and Solving, we get,

$T - T_0 = \lambda g y$ …(5)

And, $T^2 = T_0^2 + \left(\frac{\lambda s' g}{2}\right)^2$ …(6)

Using these and $T = \lambda g s$, we get,

$$S = \frac{\ell^2 + 4h^2}{8h} \quad (s' = \ell,\ y = -h)$$

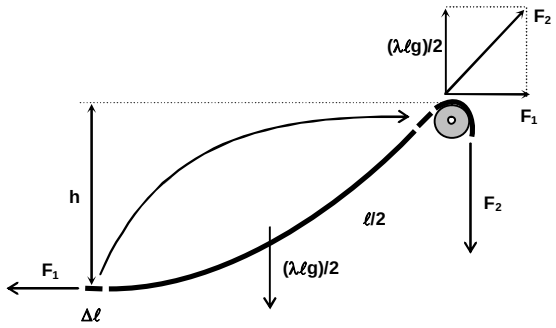
Alternative Solution:

It can be shown that the difference between the tensions in the rope at two arbitrary points on it (denoted by A and B), e.g. the centre and the right-hand pulley, depends only on the height difference between the points and the linear mass density λ of the rope. In an obvious notation: $F_B - F_A = \lambda g(h_B - h_A)$.

To prove this, consider the energy changes that would be involved if a small length of rope were (notionally) cut out from the neighbourhood of point A and inserted close to point B.

Any arbitrary piece of the rope is in equilibrium, and so the horizontal component of the tension in the rope between the pulleys is constant (equal to F_1, say). The vertical component changes from point to point; it is zero in the middle, and at the pulleys is equal to half the weight of the 'sagging' section, namely $\lambda(\ell/2)g$, where λ is the linear mass density of the rope.

As illustrated in figure, we imagine a small piece of the rope, of length $\Delta\ell$, cut out from the middle of the system, and re-inserted into the rope near one of the pulleys! What energy changes are involved?



Closing the gap in the middle of the rope requires work $F_1 \Delta l$ to be done, while the work required to lift the small piece against gravity is $\lambda\, \Delta l\, gh$. Inserting it at the pulley actually allows some energy to be recovered, but formally the work required to do this is $-F_2\, \Delta l$.

As a result of these changes, we have done no more than return the rope to its initial state! It follows that the total work done must be zero, and so

$F_1\, \Delta l + \lambda\, \Delta l\, gh - F_2\, \Delta l = 0$,

showing that

$F_2 - F_1 = \lambda gh$.

The same thought experiment could have been carried out for any arbitrary pair of points on the rope, and so it is generally true that 'the difference in tension forces at two arbitrary points of the rope is directly proportional to the height difference between those points'.

As can be seen in Fig. 1, the connection between the rope tension F2 (at the pulley) and its components is

$$F_2^{\,2} = F_1^{\,2} + \left(\frac{\lambda\, lg}{2}\right)^2$$

A third equation, one that involves s, can be established by recognising that the hanging rope segments are each held in place by a force of strength F_2:

$F_2 = \lambda gs$.

From equations (1) and (3), we have that $F_1 = \lambda g(s - h)$. Substituting this into (2) gives the final result for the length of the hanging segments as

$$s = \frac{l^2 + 4h^2}{8h}$$

Chemistry

PART – II

SECTION – A

19. Increase in concentration of the reactant shifts the equilibrium in the forward direction and vice versa.

20. 2-Ethylanthraquinol on oxidation in air produces H_2O_2.
Acetanilide retards the decomposition of H_2O_2.

21. SF_3Cl – Seesaw
In PCl_2F_3, two F atoms are in axial position and one F atom in equatorial position.

22. $B_3N_3H_6 + 9H_2O \longrightarrow 3NH_3 + 3H_3BO_3 + 3H_2$

23. General formula of cyclic silicate is $(SiO_3)_n^{2n-}$

24. The correct order are
Li < K < Na < Rb (Density)
$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2$ (Basic nature)

25. sp^3 – F_2SeO, NF_4^+,
sp^2 = NO_3^-
sp^3d = SF_2Cl_2
sp^3d^2 = IOF_4^-, XeF_5^+

26. (P) 10 ml 0.1 M H_2SO_4 + 40 ml 0.1 M NH_4OH
pOH = pK_b + log $\frac{2/50}{2/50} = 4.74$
pH = 9.26 basic buffer solution

(Q) 20 ml 0.1 M CH_3COOH + 16 ml 0.1 M NaOH
$pH = 4.74 + \log\frac{1.6}{0.4} = 5.34$
Acidic buffer solution.

(R) 50 ml 0.1 M CH_3COOH + 50 ml 0.1 M NaOH
pH = $7 + \frac{4.74}{2} + \frac{1}{2}\log\frac{5}{100} = 8.72$

(S) 10 ml 0.1 M NH_4OH + 20 ml 0.1 M NH_4Cl
pOH = $4.74 + \log\frac{2}{1} = 5.04$
pH = 14 – 5.04 = 8.96

27. P → Order of the reaction $= \frac{1}{2}$
Q → 1st order reaction.
R → zero order reaction
S → 2nd order reaction.

28. $Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$

$Mg_2C_3 + 4H_2O \longrightarrow 2Mg(OH)_2 + C_3H_4$

$CaC_2 + 4H_2O \longrightarrow Ca(OH)_2 + C_2H_4$

SECTION – D

29. $t = \frac{2.303}{k}\log\frac{a}{a-x}$

$= \frac{2.303}{5\times10^{-2}}\log\frac{20}{2}$

= 46.06

30. $K_p = P_{CO_2}$

$n = \frac{0.3\times12.315}{0.0821\times1000} = 0.045$ (n = number of moles of CO_2 at equilibrium)

$\therefore$ Mole of $CaCO_3$ = 0.2 – 0.045 = 0.155

Wt. of $CaCO_3$ = 0.155 × 100 = 15.50

31. $\underset{5}{\underset{0}{2HCl}} + \underset{5}{Ba(OH)_2} \longrightarrow \underset{0}{\underset{2.5}{BaCl_2}} + \underset{0}{\underset{5}{2H_2O}}$

$\underset{10}{\underset{5}{2CH_3COOH}} + \underset{2.5}{Ba(OH)_2} \longrightarrow \underset{0}{(CH_3COO)_2Ba} + \underset{2.5}{2H_2O}$

$pH = pK_a + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$

$pH = 4.74 + \log\frac{5}{5} = 4.74$

32. $\Delta x = \frac{h}{4\pi m\Delta V}$

$= \frac{6.625\times10^{-27}}{4\times3.14\times10^{-12}\times10^{-8}}$

$= 0.527\times10^{-7} = 5.27\times10^{-8}$ cm

= 5.27 A^o

33. Mol. Wt. of MCl_x = At. Wt. of M + 35.5x

2 × 85.02 = (Eq. wt. × x) + 35.5x

$x = \frac{2\times85.02}{7.01+35.5} = 4$

$\therefore$ At. Wt. = 7.01 × 4 = 28.04

34. $\frac{1}{\lambda} = R\times Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{1}{\lambda} = R\times4\times\left[\frac{1}{2^2} - \frac{1}{4^2}\right]$

$$\frac{1}{\lambda} = R \times 4\left[\frac{3}{16}\right]$$

$$\lambda = \frac{1}{R} \times \frac{4}{3} = 911.7 \times \frac{4}{3} A^o$$

$$\lambda = 1215.60 A^o$$

35. $t = \frac{2.303 \times 34.65}{0.693} \log \frac{100}{10}$

= 115.15 min.

36. $\Delta G^o = -2.303 RT \log K_P$

= - 2.303 × 2 × 300 × log 10^{-2}

= 2763.60 cal

Mathematics

PART – III

SECTION – A

37. We may assume m > n $\Rightarrow \lim\limits_{x\to 0} \frac{\sqrt[mn]{\cos^n x} - \sqrt[mn]{\cos^m x}}{x^2}$

$$\lim_{x\to 0} \frac{\cos^n x - \cos^m x}{x^2\left(\sqrt[mn]{(\cos^n x)^{mn-1}} + ... + \sqrt[mn]{(\cos^m x)^{mn-1}}\right)} \Rightarrow \lim_{x\to 0} \frac{\cos^n x\left(1-\cos^{m-n} x\right)}{mn x^2}$$

$$\Rightarrow \lim_{x\to 0} \frac{1-\cos^{m-n} x}{mn x^2} = \frac{(1-\cos x)\left(1+\cos x + ... + \cos^{m-n-1}\right)}{mn\, x^2} \Rightarrow \frac{m-n}{mn} \cdot \lim_{x\to\infty} \frac{1-\cos x}{x^2} = \frac{m-n}{2mn}$$

38. $g'(x) = \frac{(x-1)\left(x^3 - 3x^2 + 5x + 1\right)e^x}{\left(x^2+1\right)^3}$

Now, $x^3 - 3x^2 + 5x + 1$ is strictly increasing and has a root in (–1, 0)

39. $$g(x) = \begin{cases} f(x) & ; \ -2 < x < -1 \\ f(-1) & ; \ -1 \le x < 0 \\ f(0) & ; \ 0 \le x \le 1 \\ f(x) & ; \ x > 2 \end{cases}$$

40. We see that f(1, 1) = 1 and f(1, 2) = $1 + \frac{f(2,1)}{2}$ and $f(2,1) = 1 + \frac{f(1,2)}{2}$

So, f(1, 2) = f(2, 1) = 2, then f(3, 1) = f(1, 3) = 3

$\Rightarrow$ f(x, y) = x·y; xy $\ge$ 0

41. Since the slope of tangent is 1

We have $\frac{1+x^2}{\left(1-x^2\right)^2} = \frac{dy}{dx} = 1 \Rightarrow x^4 - 3x^2 = 0 \Rightarrow x = 0, \pm\sqrt{3}$

Now, x = 0, y = 0; x = $\sqrt{3}$, y = $\frac{-\sqrt{3}}{2}$; x = $-\sqrt{3}$, y = $\frac{\sqrt{3}}{2}$

42. Given f(x = y) = f(x) + f(y) + xyf(x + y)
Put x = y = 0 $\Rightarrow$ f(0) = 0
Put y = –x $\Rightarrow$ f(–x) = –f(x) $\Rightarrow$ f(x) is odd
and $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\left[\frac{f(h)}{h} + x(x+h)\right]$ $\because$ f(0) = 0
$\Rightarrow$ f′(0) + x^2
f′(x) = $x^2 - 1$ $\Rightarrow$ f is differentiable f′(3) = 8 and $f'(x) + f'\left(\frac{1}{x}\right) = -f'(x)\cdot f'\left(\frac{1}{x}\right)$ $\forall\ x \neq 0$

43. (P) If the $\lim_{x\to\infty} x\cdot f'(x)$ exist

So, the $\lim_{x\to\infty} \left(x\cdot f(x)\right)'$
$\Rightarrow x \to \infty$ f(x) + $\lim_{x\to\infty} x\cdot f'(x)$

$\Rightarrow \lim_{x\to\infty} \left(x\cdot f'(x)\right)' = \lim_{x\to\infty} \frac{\left(xf(x)\right)'}{x'} = \lim_{x\to\infty} \frac{x\cdot f(x)}{x} = \lim_{x\to\infty} f(x)$
So, $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} f(x) + \lim_{x\to\infty} x\cdot f'(x) = 0$

(Q) $\lim_{x\to a} \frac{e^{f(x)} - e^{g(x)}}{f(x)-g(x)}$; $\lim_{x\to\infty} e^{g(x)} \frac{\left[e^{f(x)-g(x)} - 1\right]}{f(x)-g(x)}$ use L.H. Rule

$e^{g(a)} \frac{e^{(f(x)-g(1))} - \left(f'(x) - g'(x)\right)}{\left(f'(x)-g'(x)\right)}$; $e^{g(a)} = e^{\ln 2}$ = 2

(R) Replacing f by f if necessary, we may assume f(b) > f(c) hence f(a) > f(c)
So μ be an absolute minimum of f on [a, b] which exists became the function is continuous then $\mu \in$ (a, b) and therefore f′(μ) = 0

(S) $I = \int_a^b \frac{e^{\frac{x}{a}} - e^{\frac{b}{x}}}{x} dx$ put $t = \frac{ab}{x}$ $\Rightarrow$ $x = \frac{ab}{t}$ we have $dx = \frac{ab}{t^2} dt$

$I = \int_b^a \frac{e^{\frac{ab}{at}} - e^{\frac{bt}{ab}}}{\frac{ab}{t}} \cdot \left(-\frac{ab}{t^2}\right) dt$; $I = \int_b^a \frac{e^{\frac{b}{t}} - e^{\frac{t}{a}}}{t}(-dt) = \int_a^b \frac{e^{\frac{b}{t}} - e^{\frac{t}{a}}}{t}\cdot dt = -I$ $\therefore$ I = 0

44. (P) Let $\int_{-\pi}^{\pi} \frac{x^2}{1+\sin x + \sqrt{1+\sin^2 x}} dx$ (replace x by –x)

$I = \int_{-\pi}^{\pi} \frac{x^2}{1-\sin x + \sqrt{1+\sin^2 x}} dx$; $2I = \int_{-\pi}^{\pi} x^2 \left[\frac{1}{1+\sin x + \sqrt{1+\sin^2 x}} + \frac{1}{1-\sin x + \sqrt{1+\sin^2 x}}\right] dx$

= $\int_{-\pi}^{\pi} x^2 dx$; $2I = 2\int_0^{\pi} x^2 dx$; $I = \frac{\pi^3}{3}$

(Q) Let $y = \frac{1}{x}$, $dy = -\frac{1}{x^2} dx$ then $\frac{dx}{x^2 - x + 1} = -\frac{dy}{y^2 - y + 1}$; $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{2}$; $\tan^{-1}\frac{1}{x}$; $\cot^{-1} x$

$$\therefore \frac{1}{2}\left(\int_{1/2}^{2}\frac{\tan^{-1}x}{x^2-x+1}dx+\int_{1/2}^{2}\frac{\frac{\pi}{2}-\tan^{-1}y}{y^2-y+1}\right)=\frac{\pi}{4}\int_{1/2}^{2}\frac{dx}{x^2-x+1}=\frac{\pi^2\sqrt{3}}{18}$$

(R) $\int_0^{\pi/2}\frac{\frac{\cos x}{x}-\frac{\sin x}{x^2}}{1+\left(\frac{\sin x}{x}\right)^2}dx$; Put $\frac{\sin x}{x}=t$

$\frac{x\cos x-\sin x}{x^2}dx-dt$; $\int_1^{2/\pi}\frac{dt}{1+t^2}$; $\left[\tan^{-1}x\right]_1^{2/\pi}$; $\tan^{-1}\frac{2}{\pi}-\frac{\pi}{4}$

(S) As function is odd

45. (P) $|e^x| = (0, \infty)\ \forall\ x \in R$; $\{|e^x|\} \in [0, 1)$; $[\{|e^x|\}] \Rightarrow f(x)$ is constant i.e. many one

(Q) $f(x)=e^{\text{sgn}(x)}+e^{x^2}$;

$x = 0$ $\quad f(0) = 2$

$x > 0$ $\quad f(x) = e + e^{x^2}$

$x < 0$ $\quad f(x) = \frac{1}{e}+e^{x^2}$

$\Rightarrow f(x)$ is many one

46. (P) $f(x) = x - x^2 + \sin^{-1}x$, $x \in [-1, 1]$

$f(x) \le 0$, $\forall\ x \in [-1, 0]$, $f(x) \ge 0\ \forall\ x \in [0, 1]$

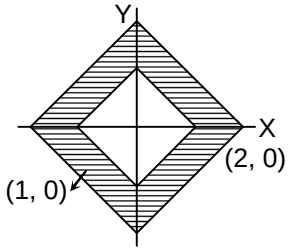
$A=-\int_{-1}^{0}\left(\sin^{-1}x+x-x^2\right)dx+\int_0^1\left(\sin^{-1}x+x-x^2\right)dx$

$= \pi - 1$

(Q) Area will be minimum if (0, 2) is mid-point of chord

(R) $1 \le |X| + |Y| \le 2$

$A=4\left(2-\frac{1}{2}\right)=6$



(S) $A=\int_0^{\frac{k}{k^2+1}}\left(x-kx^2+\frac{x^2}{k}\right)dx=\frac{k^2}{6\left(k^2+1\right)^2}=\frac{1}{6\left(k^2+\frac{1}{k^2}+2\right)}$

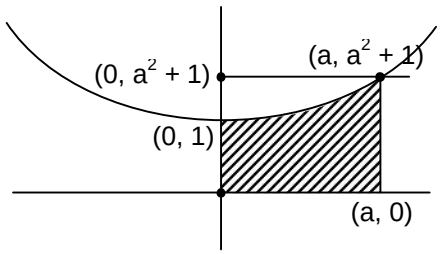
By A.M. $\ge$ G.M. equality holds $k = 1$

SECTION – D

47. Taking log $-\frac{1}{2}\left(1+\frac{1}{n}\right)\ln n+\frac{1}{n^2}\cdot\sum_{k=1}^{n}k\ln k \Rightarrow \int_0^1 x\ln x=-\frac{1}{4}$

$L = e^{-\frac{1}{4}}$

48. Area of rectangle is $a(a^2 + 1) = a^3 + a$ the portion under the parabola has area $\int_0^a \left(x^2+1\right)dx = \frac{a^3}{3} + a$



So, $\frac{\left(a^3+a\right)}{2} = \frac{a^3}{3} + a$

$\Rightarrow a = \sqrt{3}$

49. Let g: A $\to$ A be defined by $g(x) = 1-\frac{1}{x}$, then $g(g(g(x))) = 1-\frac{1}{1-\frac{1}{1-\frac{1}{x}}} = x$

So, given equation rewrites as $f(x) + f(g(x)) = \log|x|$ ..... (1)

Now, put $x = g(y)$

$f(g(y)) + f(g(g(y))) = \log|g(y)|$ ..... (2)

Again put in equation (1) $x = (g(g(z)))$

$f(g(g(z))) + (f(g(g(g(z)))) = \log|(g(g(z)))|$ ..... (3)

Now, set $y = z = x$ and by solving these equation (1), (2) and (3), we get

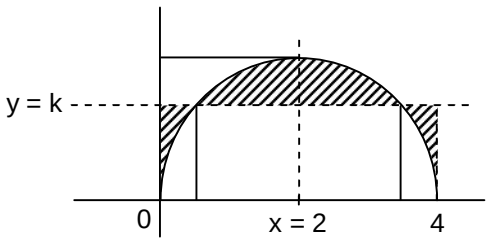
$f(x) = \frac{1}{2}\left(\log|x| - \log\left|g(x) + \log\left|\left(g\left(g(x)\right)\right)\right|\right|\right)$ put x = 2007, we have

$g(x) = \frac{2006}{2007}, g\left(g(x)\right) = -\frac{1}{2006}$

So, that $f(2007) = \log\left(\frac{2007}{2006}\right)$

50. Let $f(x) = x(4 - x)$, and $y = k$

Let A be the region below $y = f(x)$ and above $y = k$ and B be the region below $y = k$ and above $y = f(x)$ then $F(k) = A + B$



$A = 2\left(\int_0^k f^{-1}(x)dx + \int_k^4 2 - f^{-1}(x)dx\right)$

$\frac{dA}{dk} = 2\left(f^{-1}(k) - 2 + f^{-1}(k)\right), \frac{dA}{dk} = 0$

$f^{-1}(k) = 1 \Rightarrow k = 3$

51. Clearly 0, 2 are fixed points of f and therefore solution.

On the other hand, then is no solution for $x < 0$, since f is non-negative valued

For $0 < x < 2$, we have $0 \le f(x) < x < 2$ and $f(0) = 0$

For $x > 2$, $f(x) > x$ so there is no other solution

52. Let $h(x) = x(1 + \ln(x))$; h: $[1, \infty) \to [1, \infty)$

h(x) is strictly increasing and $h(1) = 1$, $\lim_{x\to\infty} h(x) \to \infty$

Hence, h(x) is Bijective

So inverse of h(x) clearly f: $[1, \infty) \to [1, \infty)$; $\lambda \to f(\lambda)$

Now, $f'(x) = \frac{1}{2 + \ln\left(f(\lambda)\right)}$

By using L. Hospital rule $\lim_{\lambda\to\infty} \frac{f(\lambda)}{\frac{\lambda}{\ln\lambda}} = \lim_{\lambda\to\infty} \frac{\frac{1}{f(\lambda)}}{\frac{1}{\lambda}\cdot\frac{1}{2+\ln(\lambda)}}$

$$= \lim_{\lambda\to\infty} \frac{f(\lambda)}{\lambda}\cdot\left(2+\ln f(x)\right) = \lim_{\lambda\to\infty} \frac{2+\ln f(\lambda)}{1+\ln f(\lambda)} = 1$$

53. $\int\left(x^5+x^2\right)\left(x^6+2x^3\right)^{\frac{1}{3}}dx$; $x^6 + 2x^3 = t$; $(6x^5 + 6x^2)\,dx = dt$

$$\int t^{\frac{1}{3}}\cdot\frac{dt}{6} = \frac{1}{6}\cdot\frac{3}{4}t^{\frac{4}{3}}+c = \frac{1}{8}\left(x^6+2x^3\right)^{\frac{4}{3}}+c$$

54. $F(x) = -\cos(\sin(\sin(\sin(x))))$
For integration put $\theta = \sin(\sin(\sin(x)))$

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